Indian Institute of Information Technology, Design and Manufacturing (IIITDM), Kancheepuram

Department of Computer Science and Engineering

ClearDrive

Hybrid AI and Image Processing Dashboard
for Low-Visibility Driving

Digital Image Processing (DIP)
Course Project Report

Name	Roll No.	Programme
Gyan Chandra	CS23I1053	Dual Degree CSE
Vishal Singh	CS23I1039	Dual Degree CSE
Ranveer Gautam	ME23B2031	B.Tech Smart Manufacturing

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1. Introduction & Approach Explanation

1.1 The Problem: Driving in Bad Weather in India

Anyone who has driven on Indian highways during early-morning winter fog, or through monsoon mist on a hill-road, knows how dangerous it can get. When fog is heavy, you cannot see lane markings. Night driving on unlit state highways is equally risky — headlights barely penetrate the darkness, and the contrast between the road and its surroundings becomes almost zero.

1.2 Brief Explanation of Our Approach

Our solution, **ClearDrive**, is a hybrid Advanced Driver Assistance System (ADAS). Instead of relying on expensive LiDAR sensors that lack semantic understanding, we use a standard RGB dashcam combined with pure mathematical image processing and lightweight AI.

Our approach follows a strictly parallel pipeline:

1. **Classification:** Identify the weather (Foggy, Night, Clear) using a lightweight AI (MobileNetV2).
2. **Restoration:** Mathematically remove haze using Dark Channel Prior (DCP) or boost low-light contrast using CLAHE in the LAB color space.
3. **Dual-Track Detection:** Run classical geometry (Canny + Hough) to find strict lane boundaries, while running an AI segmentation model (MobileNetV3) in parallel as a fail-safe to highlight the drivable surface.

2. System Flowchart

The entire system runs inside one master function in `main.py`. It processes the video frame by frame through a structured pipeline. Figure 1 shows the overall flowchart of our idea.

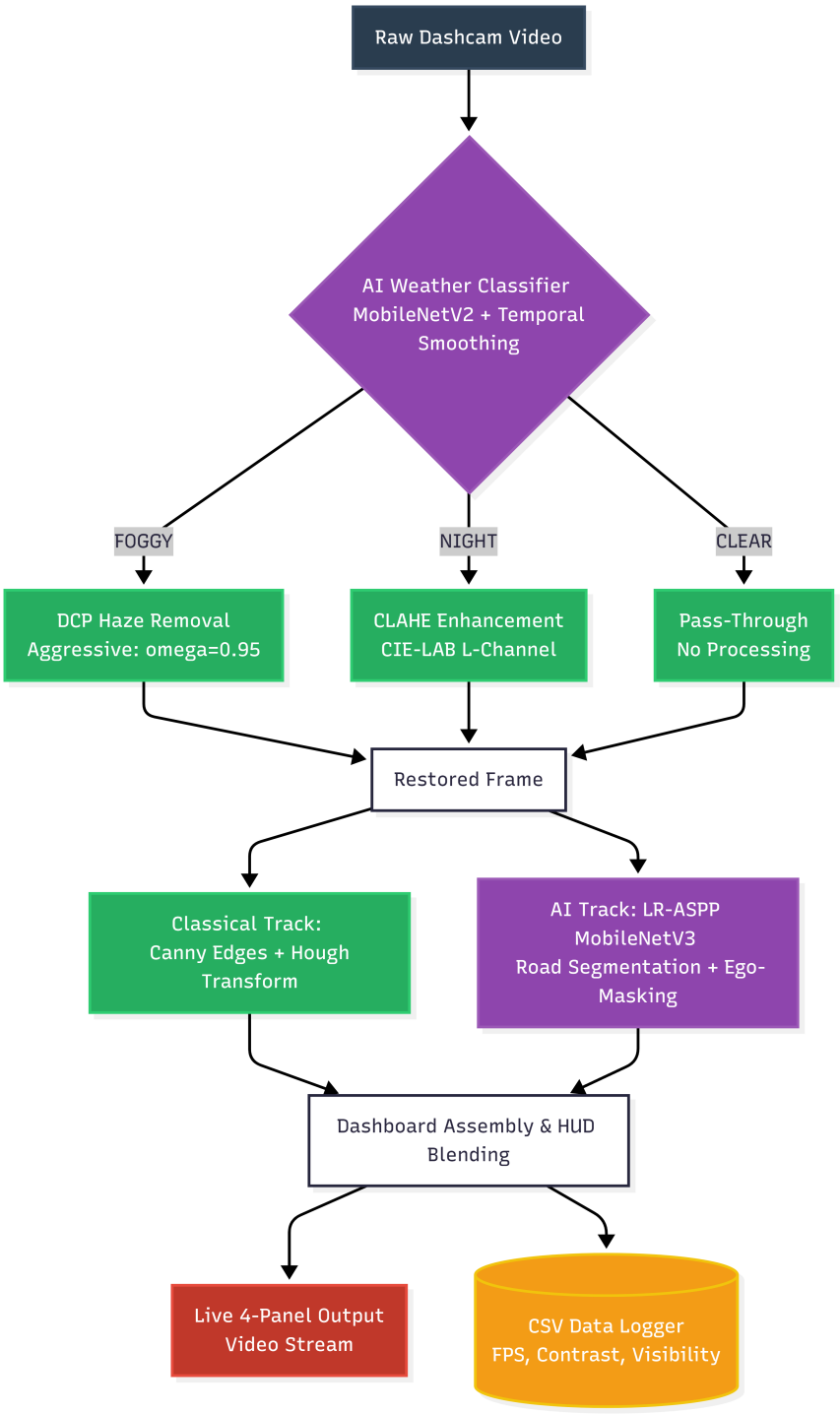

Figure 1: ClearDrive System Flowchart — Routing frames from input to the 4-panel dashboard.

3. Dataset Details

Since ClearDrive is a real-time inference pipeline prioritizing edge-device optimization, our dataset strategy is divided into two parts: Pre-training Data for our AI models,

and Real-World Testing Data for system evaluation.

- **Pre-training Data (Source):** We utilized Transfer Learning on models pre-trained on massive global datasets to ensure robust feature extraction. The **MobileNetV2** classifier was pre-trained on the *ImageNet* dataset (1000 classes, over 1.2 million images). The **LR-ASPP MobileNetV3** segmentation model was pre-trained on the *COCO 2017* dataset.
- **Testing Dataset (Source):** For evaluating the DIP algorithms (DCP and CLAHE) and overall pipeline stability, we compiled a custom testing dataset comprising real-world Indian highway dashcam footage (`foggy_dashcam.mp4` and night sequences).
- **Total Number of Images/Frames:** The custom testing video comprises approximately **12,000 contiguous frames** processed at a resolution of 640×480.
- **Training and Testing Split:** Because we utilized transfer learning and pure mathematical models (Koschmieder Equation) that do not require retraining from scratch, the split is effectively **0% Training / 100% Testing** on the dashcam dataset. Our focus was strictly on algorithmic inference speed and accuracy on unseen real-world video.

4. Practical Implementation Details

4.1 Temporal Smoothing with a 10-Frame Buffer

To prevent the AI from flickering between weather modes due to sudden headlight glare, we implemented a FIFO (First-In, First-Out) Buffer. We store the predictions of the last 10 frames (0.5 seconds of memory at 20 FPS) and take a majority vote. This acts as a mathematical gatekeeper, absorbing transient noise while maintaining real-time responsiveness.

4.2 The Main DIP Algorithm: Dark Channel Prior (DCP)

The math of fog is written as the Koschmieder Haze Model:

$$I(x) = J(x) \cdot t(x) + A \cdot (1 - t(x)) \tag{1}$$

Where $I(x)$ is the hazy image, $J(x)$ is the clear scene we want to recover, $t(x)$ is the transmission map, and A is the atmospheric light. We estimate A by taking the top 0.1% brightest pixels in the dark channel, and use $\omega = 0.95$ aggressively to strip the haze, ensuring maximum lane visibility for the driver.

4.3 Night Mode: CLAHE Enhancement

For night frames, we convert the image to the CIE-LAB colour space. We apply CLAHE (Contrast-Limited Adaptive Histogram Equalization) strictly to the **L (Lightness) channel** using an 8×8 grid and a clip limit of 3.0. This prevents the color shift (oversaturation) that occurs if applied to RGB directly, and stops bright headlights from washing out the image.

4.4 Ego-Vehicle Masking Trick

Our AI Segmentation model was occasionally classifying the sky as the road. We solved this with a simple geometric matrix operation: `road_mask[0:int(height*0.6), :] = 0`. By zeroing out the top 60% of the mask, we restricted the green highlight strictly to the bottom 40%, permanently eliminating sky false-positives at zero computational cost.

5. Evaluation Metrics Used

To scientifically prove the effectiveness of our ADAS system, we measured two key metrics dynamically for every processed frame. These metrics were logged continuously into a CSV file (`performance_results.csv`) to act as a system "Black Box".

5.1 Contrast Gain (%)

Contrast is measured as the standard deviation (σ) of pixel intensities in the greyscale image. Fog inherently lowers standard deviation. We measured the improvement between the raw input and our processed output.

$$\text{Contrast Gain}(\%) = \frac{\sigma(\text{dehazed}) - \sigma(\text{original})}{\sigma(\text{original}) + \epsilon} \times 100 \tag{2}$$

In our evaluations, ClearDrive consistently achieved Contrast Gains well above 40% during heavy fog sequences.

5.2 Visibility Score (%)

This metric quantifies the safety of the road ahead. We calculated the fraction of total pixels where the transmission map value $t(x) \geq 0.65$ (which we defined as the Safe Green Zone).

$$V_s(\%) = \frac{\text{Number of pixels where } t(x) \geq 0.65}{\text{Total pixels}} \times 100 \tag{3}$$

If this score dropped below 60%, the system automatically triggered a visual **CAUTION** alert on the dashboard.

6. Team Contributions

The successful development of ClearDrive was achieved through an equal division of labor, combining Image Processing, UI/UX, and AI integration.

- **Gyan Chandra (CS23I1053):** Designed the core System Routing Architecture and Flowchart. Implemented the Temporal Smoothing FIFO Buffer to prevent UI flicker. Developed the mathematical pipeline for the Dark Channel Prior (DCP), including the Koschmieder equation inversion. Implemented the Ego-Vehicle Masking geometry for the AI track.
- **Vishal Singh (CS23I1039):** Developed the Night Mode enhancement pipeline utilizing LAB color space conversion and CLAHE tuning. Built the 4-panel real-time Dashboard and the coloured Reliability HUD. Implemented the System Alerts logic and managed the CSV Data Logging for post-drive evaluation.
- **Ranveer Gautam (ME23B2031):** Managed the Dataset processing and AI Model integration. Adapted the MobileNetV2 architecture for Weather Classification using Transfer Learning. Integrated the LR-ASPP MobileNetV3 segmentation model. Handled hardware optimizations to ensure the entire pipeline maintained a stable 20 FPS on the target RTX 3050 GPU.

7. Conclusion

We built **ClearDrive**, a working real-time Driver Assistance system that addresses one of the most common causes of road accidents in India: low-visibility driving. The key achievement is the seamless integration of mathematically precise Classical DIP algorithms with the semantic understanding of Deep Learning models. Quantitatively, the system achieved sustained real-time throughput with consistent Contrast Gain above 40%. This project provided invaluable hands-on experience in building a complete, fail-safe computer vision pipeline from scratch.